Space for rough work

8. In which of the following elements, is the energy required to remove an electron from a neutral atom in its gaseous states the least?

(a) K (b) Si (c) P (d) Na

9. **Assertion (A):** Elements with greater electron affinities are better oxidizing agents.

Reason (R): Oxidation is gaining of electron.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

10. Non-metallic character is _______.

(a) inversely proportional to the number of valence electrons
(b) directly proportional to the number of protons
(c) directly proportional to electronegativity
(d) inversely proportional to electron affinity

11. Which of the following ionization energies has the maximum difference in a sodium atom?

(a) 2nd and 1st
(b) 3rd and 2nd
(c) 4th and 3rd
(d) In all cases, the difference must be equal.

12. Which of the following species has the largest radius?

(a) O^{-2} (b) F^{-} (c) Na^{+} (d) Mg^{+2}

13. Reducing character is directly proportional to the __________.

(a) non-metallic character (b) electronegativity
(c) tendency to lose electrons (d) tendency to gain electrons

14. A non-metal can occupy __________.

(a) the left portion of the periodic table
(b) the right portion of the periodic table
(c) the middle potion of the periodic table
(d) any place in the periodic table depending on its atomic number

15. Match the entries of Column A with those of Column B.

Column A (Atomic Number)	Column B (Description of an Element)
(i) 14	(A) An element with the highest metallic character in the period
(ii) 37	(B) Smallest alkaline earth metal
(iii) 4	(C) Maximum oxidizing power in group 16
(iv) 8	(D) The only metalloid belonging to third period

(a) (i) → (B); (ii) → (A); (iii) → (C); (iv) → (D)
(b) (i) → (A); (ii) → (C); (iii) → (D); (iv) → (B)
(c) (i) → (D); (ii) → (A); (iii) → (B); (iv) → (C)
(d) (i) → (D); (ii) → (A); (iii) → (C); (iv) → (B)

Space for rough work

Answer Keys

Assessment Test I

1. (a) **2.** (c) **3.** (b) **4.** (a) **5.** (c) **6.** (b) **7.** (b) **8.** (a) **9.** (a) **10.** (c)
11. (b) **12.** (d) **13.** (d) **14.** (c) **15.** (d)

Assessment Test II

1. (c) **2.** (c) **3.** (d) **4.** (d) **5.** (b) **6.** (a) **7.** (a) **8.** (a) **9.** (b) **10.** (b)
11. (b) **12.** (a) **13.** (a) **14.** (a) **15.** (a)

Assessment Test III

1. (a) **2.** (d) **3.** (b) **4.** (c) **5.** (a) **6.** (a) **7.** (d) **8.** (a) **9.** (a) **10.** (a)
11. (b) **12.** (a) **13.** (d) **14.** (c) **15.** (b)

Assessment Test IV

1. (b) **2.** (b) **3.** (c) **4.** (a) **5.** (b) **6.** (b) **7.** (b) **8.** (a) **9.** (c) **10.** (c)
11. (a) **12.** (a) **13.** (c) **14.** (b) **15.** (c)

Chemical Bonding 4

Reference: Coursebook - IIT Foundation Chemistry Class 9; Chapter - Chemical Bonding; Page number - 4.1–4.11

Assessment Test I

Time: 30 min.

Directions for questions from 1 to 15: Select the correct answer from the given options.

Space for rough work

1. Which of the following is not associated with an ionic bond?
   (a) $NaCl$ (b) $MgCl_2$
   (c) $AlCl_3$ (d) $CaCl_2$

2. Which of the following forces of attraction acts among the atoms in a molecule of type B – A – B, where A and B belong to group 16 and group 17, respectively?
   (a) Dipole–dipole interaction
   (b) Hydrogen bonding
   (c) Van der Waals force
   (d) Interionic attractions

3. Find the odd one out based on the presence of dative bond.
   (a) Ammonia (b) Water
   (c) Methane (d) Hydrochloric acid

4. In which of the following molecules does the atom, which is underlined, not attain octet configuration?
   (a) $\underline{N}H_3$ (b) $\underline{P}Cl_3$
   (c) $\underline{B}Cl_3$ (d) $\underline{C}HCl_3$

5. Which of the following characteristics of elements is the most suitable for ionic bond formation?
   (a) Significant difference in ionization energy
   (b) Bigger size of cation and anion
   (c) Smaller size of cation and anion
   (d) Less difference in number of valence electrons

6. Which of the following signifies oxidation?
   (a) $A^+ \rightarrow A^{++}$ (b) $A^+ \rightarrow A$
   (c) $A \rightarrow A^-$ (d) $A^{++} \rightarrow A^+$

Space for rough work

7. Which of the following is not the effect of intermolecular H-bonding?
   (a) Ice floats on water.
   (b) Ammonia gas is easily liquefiable.
   (c) Dimerization of hydrogen fluoride molecules
   (d) Co-existence of water and water vapour in nature

8. Which of the following is a redox reaction?
   (a) $Fe + S \rightarrow FeS$
   (b) $H_2O + CO_2 \rightarrow H_2CO_3$
   (c) $CaCO_3 \rightarrow CaO + CO_2$
   (d) $NaOH + HCl \rightarrow NaCl + H_2O$

9. Identify the bond present in CH_2Cl_2.
   (a) Polar covalent bond
   (b) Non-polar covalent bond
   (c) Dative bond
   (d) Ionic bond

10. Arrange the following in the increasing order of the strength of ionic bond.

(A) K_2O	(B) CaO	(C) MgO	(D) KCl
(a) CDBA	(b) BCDA	(c) DABC	(d) ACDB

11. Arrange the following in the ascending order of the number of covalent bonds present in them.

(A) N_2	(B) O_2	(C) H_2	(D) CO_2
(a) ACBD	(b) DABC	(c) BCDA	(d) CBAD

12. **Assertion (A):** Metallic bond in calcium is stronger than in potassium.
    Reason (R): The radii of alkaline earth metals are smaller than the corresponding alkali metals.
    (a) Both A and R are true and R is the correct explanation of A.
    (b) Both A and R are true, but R is not the correct explanation of A.
    (c) A is true and R is false.
    (d) A is false and R is true.

13. **Assertion (A):** Metals are generally good oxidizing agents.
    Reason (R): Metals are electropositive in nature.
    (a) Both A and R are true and R is the correct explanation of A.
    (b) Both A and R are true, but R is not the correct explanation of A.
    (c) A is true and R is false.
    (d) A is false and R is true.

Space for rough work

14. Match the statements of Column A with those of Column B.

Column A (Atoms)	Column B (Molecule Formed)
(i) A has 3 protons and B has 17 protons.	(A) A_2B
(ii) A has 8 protons and B has 9 protons.	(B) AB_3
(iii) A has 17 protons and B has 8 protons.	(C) AB
(iv) A has 5 protons and B has 9 protons.	(D) AB_2

(a) (i) → (A); (ii) → (B); (iii) → (D); (iii) → (C)
(b) (i) → (C); (ii) → (D); (iii) → (A); (iii) → (B)
(c) (i) → (B); (ii) → (C); (iii) → (D); (iii) → (A)
(d) (i) → (D); (ii) → (B); (iii) → (A); (iii) → (C)

15. Which of the following criteria is not pertinent to ionic compounds?
(a) They do not conduct electricity even in the molten state.
(b) Ions are the constituent particles.
(c) They have high density.
(d) They are generally soluble in water.

Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Which of the following compounds is associated with only ionic bond?
   (a) NH_4Cl (b) H_2O (c) CCl_4 (d) KCl

2. In which of the following does H-bonding exist?
   (a) A-H, where A is a metal.
   (b) H-A, where A is a halogen with three shells.
   (c) H_2A, where A is a non-metal with three shells.
   (d) AH_3, where A is a non-metal with two shells and five valence electrons.

3. Find the odd one out based on dative bond.
   (a) KCl (b) NH_4Cl
   (c) CCl_4 (d) HCl gas

4. Find the odd one out with respect to the valence electronic configuration of the central (underlined) atom.
   (a) $\underline{P}Cl_5$ (b) $\underline{Sn}Cl_4$
   (c) $\underline{Cl}F_3$ (d) $\underline{S}F_4$

5. Which of the following conditions is the most suitable for covalent bond formation?
   (a) Both the elements involved in the bond formation should be non-metals.
   (b) One of participating atoms should be a metal and the other one should be a non-metal.
   (c) One of the elements should be a good reducing agent and the other one should be a good oxidizing agent.
   (d) One of the participating atoms should be bigger in size than the other one.

6. Find the odd one out with respect to transfer of electrons.
   (a) $A \rightarrow A^+$ (b) $A^{-1} \rightarrow A^{-2}$
   (c) $A \rightarrow A^{-1}$ (d) $A^{++} \rightarrow A^+$

7. Which of the following is the effect of intermolecular H-bonding?
   (a) Existence of water in liquid state.
   (b) High volatility of alcohols.
   (c) Acids furnish H_3O^+ in water.
   (d) Ammonia forms its hydroxide when it is passed through water.

Space for rough work

8. Which of the following is not a redox reaction?
(a) $CaO + H_2O \rightarrow Ca(OH)_2$
(b) $Ca + H_2O \rightarrow Ca(OH)_2 + H_2$
(c) $Ca + HCl \rightarrow CaCl_2 + H_2$
(d) $Ca(NO_3)_2 \rightarrow CaO + NO_2 + O_2$

9. Identify the bonds present in HClO.
(a) Polar covalent bond
(b) Nonpolar covalent bond
(c) Dative bond
(d) Both (a) and (c)

10. Arrange the following in the decreasing order of the strength of ionic bond.
(A) CsCl (B) BaO (C) SrO (D) Cs_2O
(a) CBDA (b) ADBC (c) BCDA (d) CDAB

11. Arrange the following in the descending order of the number of covalent bonds present in them.
(A) C_2H_2 (B) $CHCl_3$ (C) NH_3 (D) Cl_2
(a) BADC (b) CDBA (c) ABCD (d) DCBA

12. **Assertion (A):** Sodium can be cut with a knife.
Reason (R): The bigger the size of the metal kernel, the stronger is the metallic bond.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

13. **Assertion (A):** Non-metals are generally good oxidizing agents.
Reason (R): Electron affinity decreases with increase in non-metallic character of the elements.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

14. Match the statements of Column A with those of Column B.

Column A (Description of Atoms Forming the Molecule)	**Column B (Forces of Attraction among the Molecules)**
(i) A has 7 protons and B has 1 proton.	(A) Interionic attraction
(ii) Both the atoms have identical electronic configuration.	(B) Dipole-dipole interaction
(iii) A has 20 protons and B has 8 protons.	(C) H – bonding
(iv) A has 1 proton and B has 16 protons.	(D) Van der Waals' forces of attraction

Space for rough work

(a) (i) → (C); (ii) → (D); (iii) → (A); (iv) → (B)
(b) (i) → (A); (ii) → (C); (iii) → (D); (iv) → (B)
(c) (i) → (B); (ii) → (C); (iii) → (A); (iv) → (D)
(d) (i) → (D); (ii) → (B); (iii) → (A); (iv) → (C)

15. Which of the following is a distinguishing feature between a polar and a non-polar compound?

(a) Polar compounds generally ionize in water.
(b) Polar compounds generally exist in liquid or gaseous state.
(c) Polar compounds do not conduct electricity in the pure form.
(d) Intermolecular forces of attraction are weak in polar compounds.

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** IF_7 is a stable halide of iodine.
   Reason (R): The atoms attain octet configuration during the formation of molecules.
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

2. **Assertion (A):** Ammonia has hydrogen bonding whereas HCl has no hydrogen bonding.
   Reason (R): Nitrogen has greater electronegativity than chlorine.
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

3. Arrange the following metals in the increasing order of the ease of formation of ionic chlorides.
   (A) K (B) Cs (C) Na (D) Ca
   (E) Mg
   (a) ECDAB (b) EADCB
   (c) BADCE (d) BCDAE

4. Arrange the following compounds in the increasing order of the strength of ionic bond.
   (A) KCl (B) RbCl (C) $AlCl_3$ (D) $CaCl_2$
   (a) DABC (b) CABD (c) DBAC (d) CBAD

5. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Definite geometry	(A) AlF_3
(ii) Existence in associated form	(B) HCl solution
(iii) Van der Waals forces	(C) Ethane
(iv) High melting point	(D) HF
(v) Conduction of electricity	(E) Hydrogen gas

Space for rough work

(a) (i) → (B); (ii) → (E); (iii) → (A); (iv) → (D); (v) → (C)
(b) (i) → (B); (ii) → (D); (iii) → (E); (iv) → (A); (v) → (C)
(c) (i) → (C); (ii) → (A); (iii) → (D); (iv) → (E); (v) → (B)
(d) (i) → (C); (ii) → (D); (iii) → (E); (iv) → (A); (v) → (B)

6. Identify the redox reaction among the following.
   (a) Reaction of potassium hydroxide with sulphuric acid.
   (b) Reaction of silver nitrate with sodium chloride.
   (c) Reaction of hydrogen sulphide with sulphur dioxide.
   (d) Decomposition of ammonium chloride.

7. A metal and a non-metal possess the electronic configurations 2, 8, 3 and 2, 8, 7. Identify the electrovalencies of the metal and non-metal.
   (a) 3, 7 (b) 3, 1 (c) 5, 1 (d) 1, 1

8. In the formation of a compound, a metal gets the configuration of argon and the non-metal gets the configuration of krypton. Identify the metal and non-metal.
   (a) Na, Br_2 (b) Na, Cl_2
   (c) Ca, Cl_2 (d) Ca, Br_2

9. Identify the incorrect pair among the following.
   (a) Hydrogen–zero lone pairs
   (b) Nitrogen–two lone pairs
   (c) Chlorine–three lone pairs
   (d) Oxygen–two lone pairs

10. Which of the following pairs of atomic numbers corresponds to the formation of covalent compound?
   (a) 13, 9 (b) 17, 8 (c) 18, 17 (d) 12, 16

11. Which of the following characteristics of metals is not attributed to the presence of metallic bond?
   (a) Malleable nature (b) Metallic lustre
   (c) Reducing property (d) Thermal conductivity

12. Which of the following is a non-bonded force of attraction?
   (a) Ionic bond (b) Covalent bond
   (c) Dative bond (d) Hydrogen bond

13. Which of the following molecules is associated with triple covalent bond?
   (a) Ethane (b) Ethylene
   (c) Acetylene (d) Carbon dioxide